CLONALG — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Nota sobre este algoritmo

Algoritmo clonal. CORREGIDO 2026-09-04: gastaba 3.5x el MaxFES. Estos resultados usan el presupuesto honesto, por eso son PEORES que los previos: el arreglo le quito presupuesto, no le arreglo un mecanismo.

Resumen — posicion entre los 11 algoritmos

dimension	presupuesto	rango medio de Friedman	posicion de 11	funciones donde es el mejor
D10	5k FES	9.99	11º	0 de 71
D10	50k FES	10.28	11º	0 de 71
D10	500k FES	10.34	11º	0 de 71
D50	5k FES	10.38	11º	0 de 71
D50	50k FES	10.48	11º	0 de 71
D50	500k FES	10.65	11º	0 de 71

CLONALG — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	5.86e+06	1.30e+08	7.28e+07	7.69e+07	2.89e+07	5.86e+06	5.33e+07	3.12e+07	3.05e+07	1.11e+07	5.86e+06	2.68e+07	1.37e+07	1.47e+07	5.44e+06
F22014	2014	10	3.61e+09	1.39e+10	9.24e+09	9.16e+09	2.24e+09	2.73e+09	8.20e+09	5.85e+09	5.72e+09	1.47e+09	1.70e+09	5.11e+09	3.73e+09	3.63e+09	7.75e+08
F32014	2014	10	2.01e+04	1.11e+05	6.88e+04	6.77e+04	2.14e+04	1.77e+04	6.53e+04	4.11e+04	4.03e+04	1.12e+04	9.89e+03	4.18e+04	2.56e+04	2.47e+04	7.61e+03
F42014	2014	10	3.99e+02	1.16e+03	7.71e+02	7.94e+02	2.01e+02	1.91e+02	5.80e+02	3.72e+02	3.84e+02	1.00e+02	9.97e+01	3.53e+02	2.10e+02	2.15e+02	5.06e+01
F52014	2014	10	2.03e+01	2.07e+01	2.05e+01	2.05e+01	1.06e-01	2.02e+01	2.05e+01	2.04e+01	2.04e+01	7.06e-02	2.01e+01	2.03e+01	2.02e+01	2.02e+01	4.11e-02
F62014	2014	10	9.66e+00	1.21e+01	1.08e+01	1.09e+01	6.41e-01	8.62e+00	1.06e+01	9.74e+00	9.68e+00	4.64e-01	7.22e+00	9.52e+00	8.75e+00	8.67e+00	4.85e-01
F72014	2014	10	5.75e+01	1.36e+02	1.03e+02	1.03e+02	1.68e+01	2.91e+01	8.76e+01	6.33e+01	6.32e+01	1.28e+01	2.91e+01	5.64e+01	4.19e+01	4.36e+01	6.40e+00
F82014	2014	10	5.96e+01	1.14e+02	9.97e+01	9.75e+01	1.06e+01	5.66e+01	9.06e+01	7.73e+01	7.67e+01	8.89e+00	4.62e+01	7.72e+01	6.43e+01	6.40e+01	6.51e+00
F92014	2014	10	6.55e+01	1.27e+02	1.05e+02	1.03e+02	1.23e+01	5.80e+01	1.01e+02	8.61e+01	8.49e+01	9.19e+00	5.26e+01	8.19e+01	6.77e+01	6.84e+01	7.47e+00
F102014	2014	10	1.18e+03	1.71e+03	1.51e+03	1.49e+03	1.19e+02	8.21e+02	1.40e+03	1.19e+03	1.16e+03	1.34e+02	7.09e+02	1.10e+03	9.63e+02	9.44e+02	9.13e+01
F112014	2014	10	1.11e+03	2.13e+03	1.87e+03	1.84e+03	1.85e+02	9.75e+02	1.80e+03	1.56e+03	1.52e+03	1.81e+02	5.69e+02	1.45e+03	1.25e+03	1.23e+03	1.47e+02
F122014	2014	10	7.97e-01	2.55e+00	1.87e+00	1.83e+00	4.19e-01	6.29e-01	1.71e+00	1.28e+00	1.26e+00	2.63e-01	5.04e-01	1.17e+00	8.45e-01	8.32e-01	1.69e-01
F132014	2014	10	2.47e+00	4.83e+00	3.67e+00	3.67e+00	5.08e-01	1.48e+00	3.54e+00	2.87e+00	2.79e+00	4.12e-01	1.15e+00	2.88e+00	2.11e+00	2.07e+00	4.15e-01
F142014	2014	10	1.62e+01	4.68e+01	3.46e+01	3.35e+01	7.45e+00	1.14e+01	2.65e+01	1.81e+01	1.87e+01	3.94e+00	4.35e+00	1.81e+01	1.23e+01	1.19e+01	2.72e+00
F152014	2014	10	4.67e+03	1.73e+05	5.15e+04	6.26e+04	4.30e+04	6.92e+02	6.00e+04	1.17e+04	1.40e+04	1.07e+04	5.47e+01	6.36e+03	1.05e+03	1.78e+03	1.67e+03
F162014	2014	10	3.49e+00	4.31e+00	4.13e+00	4.11e+00	1.43e-01	3.30e+00	4.02e+00	3.87e+00	3.85e+00	1.48e-01	3.13e+00	3.85e+00	3.67e+00	3.62e+00	1.49e-01
F172014	2014	10	8.83e+04	2.58e+06	7.86e+05	8.46e+05	5.32e+05	4.25e+04	6.24e+05	1.70e+05	2.00e+05	1.28e+05	9.06e+03	1.90e+05	6.53e+04	6.99e+04	3.08e+04
F182014	2014	10	9.40e+04	3.88e+07	6.04e+06	7.52e+06	7.49e+06	9.40e+04	4.48e+06	5.90e+05	8.33e+05	8.00e+05	4.24e+04	4.01e+05	1.10e+05	1.30e+05	8.06e+04
F192014	2014	10	6.07e+01	1.16e+05	1.59e+04	2.42e+04	2.60e+04	2.86e+01	6.72e+03	8.12e+02	1.24e+03	1.34e+03	1.56e+01	8.06e+02	1.30e+02	1.72e+02	1.57e+02
F202014	2014	10	2.54e+07	5.75e+09	2.93e+08	8.09e+08	1.15e+09	6.45e+06	1.62e+08	5.56e+07	5.76e+07	2.83e+07	2.28e+06	5.29e+07	1.56e+07	1.98e+07	1.28e+07
F212014	2014	10	2.17e+05	2.49e+07	6.53e+06	7.76e+06	5.93e+06	2.17e+05	5.04e+06	1.33e+06	1.36e+06	8.69e+05	4.39e+04	7.71e+05	4.10e+05	4.04e+05	1.91e+05
F222014	2014	10	9.23e+02	2.60e+05	5.37e+03	1.95e+04	4.19e+04	2.91e+02	4.08e+03	1.26e+03	1.49e+03	8.80e+02	2.91e+02	1.19e+03	8.20e+02	7.69e+02	2.30e+02
F232014	2014	10	3.36e+02	5.84e+02	3.98e+02	4.08e+02	5.14e+01	2.99e+02	3.98e+02	3.49e+02	3.47e+02	2.07e+01	2.71e+02	3.36e+02	3.06e+02	3.06e+02	1.22e+01
F242014	2014	10	1.79e+02	2.48e+02	2.23e+02	2.25e+02	1.20e+01	1.76e+02	2.20e+02	2.06e+02	2.04e+02	1.07e+01	1.74e+02	1.99e+02	1.86e+02	1.86e+02	6.08e+00
F252014	2014	10	2.04e+02	2.18e+02	2.09e+02	2.10e+02	2.83e+00	2.01e+02	2.08e+02	2.05e+02	2.05e+02	1.61e+00	1.90e+02	2.04e+02	2.02e+02	2.01e+02	2.85e+00
F262014	2014	10	1.03e+02	1.05e+02	1.04e+02	1.04e+02	4.76e-01	1.02e+02	1.04e+02	1.03e+02	1.03e+02	3.71e-01	1.01e+02	1.03e+02	1.02e+02	1.02e+02	3.70e-01
F272014	2014	10	3.51e+02	6.16e+02	4.87e+02	4.88e+02	4.73e+01	2.11e+02	4.87e+02	4.34e+02	4.06e+02	6.99e+01	9.20e+01	3.63e+02	2.28e+02	2.28e+02	6.03e+01
F282014	2014	10	4.32e+02	4.95e+02	4.58e+02	4.58e+02	1.45e+01	4.05e+02	4.43e+02	4.30e+02	4.28e+02	8.13e+00	4.01e+02	4.29e+02	4.17e+02	4.16e+02	5.86e+00
F292014	2014	10	1.06e+07	4.06e+07	2.11e+07	2.18e+07	7.08e+06	6.04e+06	1.46e+07	1.00e+07	1.00e+07	2.05e+06	4.70e+06	9.87e+06	6.81e+06	6.83e+06	8.64e+05
F302014	2014	10	2.23e+05	4.08e+07	9.27e+06	1.18e+07	1.01e+07	9.44e+04	5.83e+06	1.05e+06	1.53e+06	1.30e+06	4.69e+04	7.43e+05	2.28e+05	2.57e+05	1.56e+05
F12017	2017	10	3.33e+09	1.14e+10	8.13e+09	8.41e+09	2.08e+09	2.50e+09	6.41e+09	4.84e+09	4.84e+09	9.27e+08	1.69e+09	4.84e+09	3.11e+09	3.16e+09	7.29e+08
F22017	2017	10	1.23e+04	3.59e+04	2.32e+04	2.31e+04	6.12e+03	5.09e+03	2.00e+04	1.36e+04	1.36e+04	3.25e+03	4.43e+03	1.23e+04	7.71e+03	7.92e+03	2.03e+03
F32017	2017	10	3.00e+02	1.85e+03	1.07e+03	1.05e+03	3.94e+02	1.55e+02	8.94e+02	5.12e+02	5.07e+02	1.57e+02	9.36e+01	4.47e+02	2.66e+02	2.72e+02	8.21e+01
F42017	2017	10	5.60e+03	1.96e+04	1.38e+04	1.35e+04	3.17e+03	4.28e+03	1.11e+04	8.44e+03	8.23e+03	1.48e+03	2.77e+03	6.89e+03	5.16e+03	5.16e+03	1.04e+03
F52017	2017	10	1.26e-03	1.33e-02	5.82e-03	6.33e-03	2.37e-03	5.98e-04	4.10e-03	1.81e-03	2.04e-03	8.92e-04	1.15e-04	1.85e-03	7.39e-04	7.52e-04	3.82e-04
F62017	2017	10	1.57e+05	3.99e+05	2.88e+05	2.80e+05	6.30e+04	7.45e+04	2.28e+05	1.69e+05	1.69e+05	3.25e+04	6.28e+04	1.58e+05	1.13e+05	1.11e+05	1.90e+04
F72017	2017	10	1.28e+02	2.52e+02	1.94e+02	1.93e+02	2.74e+01	9.47e+01	1.85e+02	1.40e+02	1.41e+02	1.97e+01	7.00e+01	1.26e+02	1.03e+02	1.03e+02	1.30e+01
F82017	2017	10	2.89e+00	1.34e+01	8.56e+00	8.65e+00	2.33e+00	2.89e+00	8.81e+00	5.62e+00	5.52e+00	1.15e+00	1.58e+00	5.42e+00	3.10e+00	3.15e+00	7.90e-01
F92017	2017	10	1.53e+03	2.31e+03	1.92e+03	1.90e+03	1.76e+02	1.00e+03	1.85e+03	1.55e+03	1.52e+03	1.77e+02	9.27e+02	1.48e+03	1.30e+03	1.27e+03	1.31e+02
F102017	2017	10	1.00e+05	2.80e+07	6.73e+06	8.12e+06	6.40e+06	1.00e+05	6.10e+06	8.46e+05	1.25e+06	1.15e+06	4.19e+04	6.54e+05	2.30e+05	2.65e+05	1.57e+05
F112017	2017	10	5.84e+07	1.37e+09	5.83e+08	5.89e+08	2.99e+08	5.56e+07	4.10e+08	1.57e+08	1.79e+08	8.55e+07	1.46e+07	1.48e+08	8.37e+07	8.23e+07	2.57e+07
F122017	2017	10	2.48e+07	3.96e+08	1.11e+08	1.36e+08	8.40e+07	9.88e+05	7.12e+07	1.89e+07	2.27e+07	1.54e+07	3.52e+05	1.66e+07	5.79e+06	6.20e+06	3.60e+06
F132017	2017	10	1.17e+04	8.26e+04	4.15e+04	4.28e+04	1.66e+04	2.92e+03	4.68e+04	1.89e+04	1.96e+04	8.39e+03	2.15e+03	1.51e+04	7.57e+03	7.74e+03	3.24e+03
F142017	2017	10	1.31e+06	9.57e+06	4.04e+06	4.38e+06	1.95e+06	1.60e+05	2.27e+06	1.21e+06	1.09e+06	4.70e+05	7.01e+04	9.37e+05	3.33e+05	3.63e+05	1.86e+05
F152017	2017	10	2.82e+04	8.35e+05	1.18e+05	1.86e+05	1.72e+05	4.62e+03	1.03e+05	3.62e+04	4.05e+04	2.18e+04	3.80e+03	2.61e+04	1.44e+04	1.46e+04	5.53e+03
F162017	2017	10	4.79e+03	1.33e+05	2.58e+04	3.42e+04	2.46e+04	2.92e+03	2.04e+04	8.78e+03	9.30e+03	4.37e+03	5.74e+02	8.27e+03	4.18e+03	4.23e+03	1.60e+03
F172017	2017	10	5.18e+04	2.62e+07	4.09e+06	6.58e+06	6.33e+06	5.18e+04	3.12e+06	4.21e+05	6.86e+05	7.23e+05	3.69e+04	5.46e+05	8.99e+04	1.17e+05	9.21e+04
F182017	2017	10	1.97e+07	4.43e+09	7.39e+08	1.14e+09	1.21e+09	5.85e+04	2.80e+08	4.11e+07	5.86e+07	5.82e+07	5.45e+04	9.39e+06	1.41e+06	2.12e+06	2.14e+06
F192017	2017	10	4.55e+02	2.44e+03	1.18e+03	1.21e+03	4.32e+02	1.81e+02	9.44e+02	6.29e+02	6.31e+02	1.76e+02	1.19e+02	5.40e+02	3.28e+02	3.32e+02	1.02e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.12e+03	2.47e+03	1.75e+03	1.78e+03	3.09e+02	7.89e+02	1.54e+03	1.26e+03	1.22e+03	1.69e+02	7.19e+02	1.19e+03	9.88e+02	9.78e+02	1.02e+02
F212017	2017	10	1.28e+02	1.65e+02	1.55e+02	1.53e+02	7.38e+00	1.28e+02	1.48e+02	1.42e+02	1.41e+02	4.85e+00	1.24e+02	1.37e+02	1.31e+02	1.31e+02	2.95e+00
F222017	2017	10	1.63e+03	2.65e+03	2.14e+03	2.15e+03	2.34e+02	1.13e+03	1.95e+03	1.68e+03	1.67e+03	1.39e+02	7.52e+02	1.66e+03	1.43e+03	1.41e+03	1.58e+02
F232017	2017	10	2.97e+03	4.31e+03	3.64e+03	3.62e+03	3.07e+02	2.19e+03	3.30e+03	3.04e+03	3.01e+03	2.34e+02	1.89e+03	2.97e+03	2.73e+03	2.69e+03	2.16e+02
F242017	2017	10	6.88e+02	1.52e+03	1.04e+03	1.03e+03	1.83e+02	6.69e+02	9.99e+02	7.98e+02	8.05e+02	7.76e+01	6.00e+02	7.80e+02	6.93e+02	6.94e+02	4.03e+01
F252017	2017	10	4.74e+02	5.04e+02	4.93e+02	4.91e+02	7.94e+00	4.68e+02	4.85e+02	4.77e+02	4.78e+02	3.93e+00	4.22e+02	4.77e+02	4.69e+02	4.69e+02	7.59e+00
F262017	2017	10	4.11e+02	4.28e+02	4.17e+02	4.18e+02	3.97e+00	4.04e+02	4.14e+02	4.10e+02	4.10e+02	2.17e+00	4.01e+02	4.08e+02	4.05e+02	4.05e+02	1.78e+00
F272017	2017	10	4.61e+02	5.27e+02	5.05e+02	5.02e+02	1.06e+01	2.95e+02	4.97e+02	4.80e+02	4.63e+02	3.87e+01	2.95e+02	4.75e+02	4.30e+02	4.17e+02	4.35e+01
F282017	2017	10	2.58e+04	1.97e+07	7.13e+05	1.74e+06	3.20e+06	5.71e+03	3.19e+05	5.09e+04	7.24e+04	6.80e+04	2.97e+03	3.83e+04	1.06e+04	1.16e+04	6.56e+03
F292017	2017	10	2.27e+07	7.55e+08	3.00e+08	3.20e+08	1.75e+08	2.57e+06	1.47e+08	4.98e+07	5.72e+07	3.38e+07	2.57e+06	3.81e+07	1.77e+07	1.83e+07	8.75e+06
F12022	2022	10	8.76e+03	3.09e+04	1.69e+04	1.76e+04	5.06e+03	4.71e+03	1.66e+04	1.12e+04	1.07e+04	2.90e+03	3.24e+03	9.88e+03	6.55e+03	6.61e+03	1.72e+03
F22022	2022	10	1.56e+02	7.87e+02	4.39e+02	4.20e+02	1.39e+02	8.21e+01	3.05e+02	2.17e+02	2.12e+02	5.00e+01	7.45e+01	1.72e+02	1.26e+02	1.29e+02	2.35e+01
F32022	2022	10	1.98e-01	6.84e-01	5.26e-01	5.09e-01	1.01e-01	1.34e-01	4.30e-01	3.18e-01	3.07e-01	6.74e-02	1.08e-01	2.57e-01	1.93e-01	1.93e-01	3.41e-02
F42022	2022	10	1.11e+02	2.14e+02	1.73e+02	1.69e+02	2.34e+01	7.00e+01	1.59e+02	1.32e+02	1.29e+02	1.66e+01	4.20e+01	1.21e+02	9.53e+01	9.33e+01	1.60e+01
F52022	2022	10	3.61e+00	9.01e+00	6.23e+00	6.22e+00	1.22e+00	1.93e+00	5.89e+00	4.11e+00	4.11e+00	9.74e-01	1.16e+00	3.79e+00	2.65e+00	2.57e+00	6.01e-01
F62022	2022	10	3.93e+06	2.10e+08	7.31e+07	8.04e+07	4.76e+07	1.99e+06	4.65e+07	1.40e+07	1.60e+07	1.11e+07	3.55e+05	1.05e+07	2.99e+06	3.36e+06	2.24e+06
F72022	2022	10	2.02e+02	1.30e+03	6.72e+02	7.01e+02	2.67e+02	1.69e+02	5.89e+02	3.81e+02	3.68e+02	1.06e+02	1.11e+02	2.91e+02	1.90e+02	1.94e+02	4.38e+01
F82022	2022	10	2.30e+03	6.44e+07	2.12e+05	6.17e+06	1.58e+07	7.68e+02	1.69e+04	2.37e+03	3.29e+03	2.72e+03	3.63e+02	1.98e+03	8.82e+02	9.48e+02	3.64e+02
F92022	2022	10	4.25e+02	6.31e+02	5.16e+02	5.12e+02	4.17e+01	4.21e+02	4.73e+02	4.43e+02	4.43e+02	1.19e+01	3.18e+02	4.36e+02	4.25e+02	4.20e+02	1.83e+01
F102022	2022	10	3.04e+02	6.15e+02	3.82e+02	3.89e+02	5.09e+01	2.64e+02	3.47e+02	3.15e+02	3.17e+02	1.65e+01	2.49e+02	2.99e+02	2.82e+02	2.81e+02	1.16e+01
F112022	2022	10	7.35e+01	9.21e+02	5.38e+02	5.14e+02	2.34e+02	7.35e+01	4.86e+02	2.28e+02	2.36e+02	1.01e+02	6.00e+01	2.16e+02	1.20e+02	1.24e+02	3.44e+01
F122022	2022	10	1.72e+02	1.89e+02	1.79e+02	1.79e+02	3.38e+00	1.69e+02	1.76e+02	1.73e+02	1.73e+02	1.38e+00	1.69e+02	1.72e+02	1.71e+02	1.71e+02	8.03e-01

CLONALG — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	2.64e+09	7.21e+09	5.48e+09	5.32e+09	9.00e+08	2.64e+09	4.75e+09	3.92e+09	3.90e+09	4.74e+08	2.08e+09	3.75e+09	2.91e+09	2.93e+09	4.21e+08
F22014	2014	50	2.12e+11	3.01e+11	2.63e+11	2.61e+11	2.33e+10	1.65e+11	2.49e+11	2.25e+11	2.22e+11	1.94e+10	1.50e+11	2.16e+11	1.93e+11	1.90e+11	1.45e+10
F32014	2014	50	3.14e+05	6.15e+05	4.95e+05	4.87e+05	7.40e+04	2.42e+05	4.68e+05	3.60e+05	3.63e+05	4.65e+04	1.93e+05	3.38e+05	2.82e+05	2.81e+05	2.89e+04
F42014	2014	50	5.03e+04	1.27e+05	1.02e+05	9.99e+04	1.49e+04	4.63e+04	9.07e+04	7.09e+04	7.10e+04	1.13e+04	3.54e+04	7.03e+04	5.49e+04	5.50e+04	7.87e+03
F52014	2014	50	2.11e+01	2.14e+01	2.13e+01	2.13e+01	5.92e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	4.66e-02	2.11e+01	2.12e+01	2.11e+01	2.11e+01	3.29e-02
F62014	2014	50	7.27e+01	8.17e+01	7.89e+01	7.84e+01	1.96e+00	7.08e+01	7.81e+01	7.57e+01	7.51e+01	1.87e+00	6.77e+01	7.47e+01	7.21e+01	7.20e+01	1.65e+00
F72014	2014	50	1.59e+03	2.83e+03	2.46e+03	2.44e+03	2.58e+02	1.59e+03	2.52e+03	2.10e+03	2.10e+03	1.69e+02	1.52e+03	2.08e+03	1.86e+03	1.86e+03	1.25e+02
F82014	2014	50	7.96e+02	9.86e+02	9.12e+02	9.08e+02	4.19e+01	7.76e+02	9.11e+02	8.37e+02	8.39e+02	3.34e+01	7.09e+02	8.33e+02	7.96e+02	7.95e+02	2.46e+01
F92014	2014	50	9.45e+02	1.26e+03	1.15e+03	1.15e+03	7.34e+01	9.45e+02	1.15e+03	1.06e+03	1.05e+03	4.85e+01	8.15e+02	1.03e+03	9.62e+02	9.57e+02	4.88e+01
F102014	2014	50	1.43e+04	1.59e+04	1.50e+04	1.50e+04	3.53e+02	1.36e+04	1.48e+04	1.43e+04	1.43e+04	2.80e+02	1.22e+04	1.41e+04	1.37e+04	1.36e+04	4.00e+02
F112014	2014	50	1.32e+04	1.59e+04	1.52e+04	1.51e+04	5.30e+02	1.23e+04	1.51e+04	1.44e+04	1.43e+04	4.77e+02	1.23e+04	1.42e+04	1.37e+04	1.36e+04	3.57e+02
F122014	2014	50	3.05e+00	5.54e+00	4.82e+00	4.72e+00	5.15e-01	3.05e+00	4.54e+00	4.02e+00	3.98e+00	3.55e-01	2.82e+00	3.81e+00	3.42e+00	3.39e+00	2.25e-01
F132014	2014	50	1.06e+01	1.31e+01	1.18e+01	1.19e+01	5.72e-01	9.54e+00	1.17e+01	1.07e+01	1.06e+01	5.08e-01	8.50e+00	1.04e+01	9.68e+00	9.61e+00	4.05e-01
F142014	2014	50	5.39e+02	7.53e+02	6.78e+02	6.74e+02	5.63e+01	4.57e+02	6.67e+02	5.73e+02	5.73e+02	4.16e+01	3.64e+02	5.70e+02	4.88e+02	4.85e+02	4.47e+01
F152014	2014	50	3.31e+07	2.00e+08	1.00e+08	1.04e+08	4.04e+07	1.39e+07	1.04e+08	4.80e+07	5.16e+07	1.81e+07	1.39e+07	4.44e+07	3.08e+07	2.92e+07	7.22e+06
F162014	2014	50	2.32e+01	2.40e+01	2.38e+01	2.37e+01	1.66e-01	2.29e+01	2.36e+01	2.34e+01	2.34e+01	1.64e-01	2.26e+01	2.33e+01	2.31e+01	2.31e+01	1.51e-01
F172014	2014	50	2.39e+08	6.86e+08	4.08e+08	4.29e+08	1.17e+08	1.01e+08	3.55e+08	2.44e+08	2.34e+08	6.10e+07	7.50e+07	2.13e+08	1.42e+08	1.41e+08	3.13e+07
F182014	2014	50	7.59e+09	2.12e+10	1.67e+10	1.64e+10	3.10e+09	7.27e+09	1.49e+10	1.09e+10	1.11e+10	1.57e+09	5.08e+09	1.03e+10	7.53e+09	7.50e+09	9.61e+08
F192014	2014	50	2.04e+09	2.17e+10	1.02e+10	1.07e+10	4.43e+09	1.56e+09	9.38e+09	4.77e+09	4.61e+09	1.62e+09	6.24e+08	4.28e+09	2.28e+09	2.33e+09	8.47e+08
F202014	2014	50	2.56e+14	1.69e+16	6.86e+15	7.06e+15	4.26e+15	2.56e+14	3.91e+15	1.39e+15	1.51e+15	9.32e+14	3.43e+13	9.12e+14	3.25e+14	3.44e+14	2.12e+14
F212014	2014	50	1.86e+09	8.37e+09	4.74e+09	4.69e+09	1.46e+09	1.14e+09	3.73e+09	2.18e+09	2.39e+09	5.93e+08	6.87e+08	1.97e+09	1.49e+09	1.45e+09	3.43e+08
F222014	2014	50	2.63e+13	1.70e+15	2.93e+14	4.60e+14	4.60e+14	1.29e+12	1.87e+14	3.68e+13	4.87e+13	4.09e+13	2.38e+11	1.91e+13	6.25e+12	7.01e+12	4.49e+12
F232014	2014	50	9.83e+02	3.39e+03	2.27e+03	2.20e+03	5.34e+02	9.83e+02	2.29e+03	1.79e+03	1.75e+03	3.05e+02	8.04e+02	1.72e+03	1.34e+03	1.30e+03	1.84e+02
F242014	2014	50	8.58e+02	1.07e+03	9.72e+02	9.69e+02	5.56e+01	7.83e+02	9.69e+02	9.01e+02	9.01e+02	3.81e+01	7.34e+02	8.92e+02	8.25e+02	8.20e+02	3.40e+01
F252014	2014	50	5.07e+02	8.20e+02	6.65e+02	6.61e+02	6.99e+01	4.16e+02	6.63e+02	5.59e+02	5.46e+02	5.79e+01	3.68e+02	5.75e+02	4.81e+02	4.77e+02	4.04e+01
F262014	2014	50	1.12e+02	1.16e+02	1.13e+02	1.13e+02	8.25e-01	1.10e+02	1.12e+02	1.11e+02	1.11e+02	5.48e-01	1.09e+02	1.11e+02	1.10e+02	1.10e+02	4.72e-01
F272014	2014	50	2.35e+03	2.56e+03	2.47e+03	2.46e+03	5.06e+01	2.22e+03	2.40e+03	2.36e+03	2.35e+03	4.01e+01	2.17e+03	2.31e+03	2.28e+03	2.27e+03	3.27e+01
F282014	2014	50	4.64e+03	1.03e+04	7.83e+03	7.86e+03	1.11e+03	4.55e+03	7.20e+03	6.45e+03	6.32e+03	6.38e+02	3.75e+03	6.09e+03	5.27e+03	5.21e+03	4.72e+02
F292014	2014	50	4.47e+09	1.59e+10	1.08e+10	1.01e+10	2.82e+09	3.65e+09	8.86e+09	6.01e+09	5.93e+09	1.19e+09	2.14e+09	5.99e+09	3.95e+09	3.87e+09	7.68e+08
F302014	2014	50	4.25e+12	1.24e+15	1.17e+14	1.85e+14	2.08e+14	3.81e+11	8.47e+13	1.18e+13	1.82e+13	1.82e+13	3.81e+11	9.15e+12	2.73e+12	3.01e+12	2.12e+12
F12017	2017	50	1.86e+11	2.79e+11	2.27e+11	2.27e+11	1.99e+10	1.47e+11	2.26e+11	2.01e+11	1.96e+11	1.57e+10	1.39e+11	1.88e+11	1.69e+11	1.68e+11	1.19e+10
F22017	2017	50	2.00e+05	3.72e+05	3.14e+05	3.13e+05	3.74e+04	2.00e+05	3.15e+05	2.67e+05	2.65e+05	2.54e+04	1.69e+05	2.67e+05	2.27e+05	2.26e+05	2.02e+04
F32017	2017	50	5.36e+04	1.32e+05	1.02e+05	9.98e+04	1.55e+04	5.36e+04	9.33e+04	7.32e+04	7.29e+04	8.73e+03	3.23e+04	6.50e+04	5.41e+04	5.34e+04	7.58e+03
F42017	2017	50	1.78e+05	2.83e+05	2.44e+05	2.42e+05	2.28e+04	1.71e+05	2.43e+05	2.09e+05	2.07e+05	1.76e+04	1.34e+05	2.14e+05	1.82e+05	1.81e+05	1.52e+04
F52017	2017	50	7.37e-02	1.49e-01	1.25e-01	1.19e-01	1.79e-02	6.28e-02	1.15e-01	9.66e-02	9.35e-02	1.21e-02	5.04e-02	8.69e-02	7.35e-02	7.25e-02	9.03e-03
F62017	2017	50	7.42e+06	1.13e+07	9.55e+06	9.48e+06	8.88e+05	6.77e+06	9.09e+06	8.28e+06	8.22e+06	5.77e+05	4.22e+06	8.37e+06	7.15e+06	7.12e+06	6.66e+05
F72017	2017	50	1.83e+03	2.38e+03	2.16e+03	2.17e+03	1.09e+02	1.65e+03	2.15e+03	2.00e+03	1.98e+03	9.87e+01	1.60e+03	1.96e+03	1.80e+03	1.81e+03	8.13e+01
F82017	2017	50	1.28e+02	2.41e+02	1.95e+02	1.95e+02	2.25e+01	1.24e+02	1.91e+02	1.66e+02	1.64e+02	1.76e+01	9.07e+01	1.68e+02	1.41e+02	1.39e+02	1.31e+01
F92017	2017	50	1.40e+04	1.63e+04	1.55e+04	1.54e+04	4.48e+02	1.37e+04	1.52e+04	1.46e+04	1.46e+04	3.37e+02	1.34e+04	1.44e+04	1.39e+04	1.39e+04	2.41e+02
F102017	2017	50	1.42e+10	5.50e+10	3.55e+10	3.40e+10	1.06e+10	8.13e+09	2.91e+10	1.86e+10	1.88e+10	4.80e+09	5.27e+09	1.48e+10	1.07e+10	1.08e+10	2.34e+09
F112017	2017	50	2.52e+10	5.22e+10	3.97e+10	3.99e+10	5.89e+09	2.11e+10	3.74e+10	3.22e+10	3.11e+10	3.76e+09	1.64e+10	2.83e+10	2.33e+10	2.32e+10	2.95e+09
F122017	2017	50	9.64e+10	1.92e+11	1.46e+11	1.47e+11	2.00e+10	7.32e+10	1.32e+11	1.08e+11	1.05e+11	1.34e+10	5.42e+10	9.64e+10	7.68e+10	7.75e+10	9.95e+09
F132017	2017	50	3.05e+07	1.81e+08	9.48e+07	9.36e+07	4.20e+07	1.45e+07	6.96e+07	3.92e+07	3.99e+07	1.29e+07	7.54e+06	2.99e+07	1.69e+07	1.72e+07	5.40e+06
F142017	2017	50	1.83e+10	5.49e+10	3.89e+10	3.88e+10	7.21e+09	1.30e+10	3.17e+10	2.27e+10	2.28e+10	4.47e+09	1.06e+10	2.19e+10	1.71e+10	1.70e+10	2.84e+09
F152017	2017	50	2.09e+09	5.22e+10	3.22e+10	3.12e+10	1.07e+10	2.09e+09	1.92e+10	1.32e+10	1.24e+10	3.75e+09	2.09e+09	9.91e+09	6.40e+09	6.36e+09	1.61e+09
F162017	2017	50	1.39e+12	2.90e+14	9.45e+13	1.09e+14	8.18e+13	1.06e+12	3.96e+13	9.17e+12	1.11e+13	8.53e+12	5.44e+10	4.93e+12	1.17e+12	1.46e+12	1.08e+12
F172017	2017	50	5.90e+07	6.13e+08	3.31e+08	3.44e+08	1.44e+08	3.41e+07	2.41e+08	1.28e+08	1.29e+08	4.31e+07	9.66e+06	1.27e+08	6.52e+07	6.48e+07	2.60e+07
F182017	2017	50	1.39e+15	3.75e+16	1.09e+16	1.24e+16	8.70e+15	5.00e+13	9.15e+15	1.85e+15	2.08e+15	1.62e+15	2.31e+13	1.04e+15	3.64e+14	3.90e+14	2.89e+14
F192017	2017	50	1.64e+04	2.77e+04	2.16e+04	2.20e+04	2.73e+03	1.24e+04	2.00e+04	1.72e+04	1.69e+04	1.91e+03	9.43e+03	1.60e+04	1.36e+04	1.35e+04	1.63e+03

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	1.03e+05	2.48e+05	2.09e+05	2.05e+05	2.80e+04	6.82e+04	2.12e+05	1.75e+05	1.65e+05	3.14e+04	5.99e+04	1.56e+05	1.11e+05	1.11e+05	2.44e+04
F212017	2017	50	6.76e+03	1.47e+04	1.17e+04	1.13e+04	1.77e+03	3.69e+03	7.35e+03	5.78e+03	5.64e+03	8.21e+02	2.59e+03	5.09e+03	4.17e+03	4.04e+03	5.90e+02
F222017	2017	50	1.61e+05	2.26e+05	2.02e+05	1.99e+05	1.54e+04	1.20e+05	1.83e+05	1.52e+05	1.53e+05	1.38e+04	8.74e+04	1.45e+05	1.25e+05	1.24e+05	1.20e+04
F232017	2017	50	8.30e+04	1.44e+05	1.26e+05	1.24e+05	1.16e+04	6.09e+04	1.11e+05	9.01e+04	8.95e+04	1.21e+04	5.71e+04	8.74e+04	7.61e+04	7.43e+04	7.61e+03
F242017	2017	50	3.60e+04	8.13e+04	5.96e+04	5.93e+04	1.04e+04	2.74e+04	5.12e+04	4.29e+04	4.19e+04	5.44e+03	2.33e+04	4.09e+04	3.31e+04	3.27e+04	3.83e+03
F252017	2017	50	6.74e+03	1.82e+04	1.26e+04	1.23e+04	2.93e+03	4.44e+03	9.71e+03	7.28e+03	7.12e+03	1.19e+03	3.52e+03	6.66e+03	5.20e+03	5.16e+03	7.15e+02
F262017	2017	50	1.40e+03	2.81e+03	2.29e+03	2.28e+03	2.68e+02	1.32e+03	2.03e+03	1.73e+03	1.71e+03	1.49e+02	1.09e+03	1.70e+03	1.49e+03	1.48e+03	1.18e+02
F272017	2017	50	4.41e+03	7.74e+03	6.22e+03	6.17e+03	8.61e+02	3.30e+03	5.08e+03	4.50e+03	4.44e+03	4.01e+02	2.75e+03	4.19e+03	3.74e+03	3.68e+03	3.19e+02
F282017	2017	50	7.50e+12	2.57e+15	4.19e+14	5.96e+14	6.16e+14	2.32e+12	2.23e+14	3.15e+13	4.68e+13	4.18e+13	7.57e+11	2.20e+13	7.38e+12	8.04e+12	4.82e+12
F292017	2017	50	3.91e+13	1.17e+16	2.78e+15	3.38e+15	2.90e+15	3.91e+13	1.05e+15	3.83e+14	4.17e+14	2.82e+14	6.57e+11	2.35e+14	6.51e+13	8.04e+13	5.79e+13
F12022	2022	20	4.05e+04	8.43e+04	6.22e+04	6.15e+04	9.16e+03	3.11e+04	6.48e+04	4.57e+04	4.63e+04	6.71e+03	2.53e+04	4.20e+04	3.42e+04	3.37e+04	4.44e+03
F22022	2022	20	2.45e+03	6.93e+03	4.23e+03	4.31e+03	1.07e+03	1.74e+03	3.73e+03	2.67e+03	2.74e+03	4.55e+02	8.79e+02	2.45e+03	1.83e+03	1.83e+03	3.80e+02
F32022	2022	20	1.40e+00	2.38e+00	1.92e+00	1.92e+00	2.36e-01	9.24e-01	1.68e+00	1.38e+00	1.36e+00	1.75e-01	7.41e-01	1.31e+00	1.05e+00	1.05e+00	1.40e-01
F42022	2022	20	4.14e+02	5.85e+02	5.09e+02	5.04e+02	4.12e+01	3.26e+02	4.87e+02	4.40e+02	4.34e+02	3.46e+01	2.94e+02	4.14e+02	3.71e+02	3.67e+02	2.64e+01
F52022	2022	20	1.41e+01	3.43e+01	2.63e+01	2.61e+01	4.20e+00	1.07e+01	2.48e+01	1.89e+01	1.91e+01	3.12e+00	7.10e+00	1.81e+01	1.45e+01	1.43e+01	2.46e+00
F62022	2022	20	1.32e+09	4.80e+09	2.76e+09	2.82e+09	8.14e+08	7.16e+08	2.38e+09	1.58e+09	1.58e+09	4.50e+08	1.52e+08	1.14e+09	7.08e+08	6.98e+08	2.04e+08
F72022	2022	20	2.35e+03	5.93e+03	4.47e+03	4.38e+03	9.09e+02	1.53e+03	3.88e+03	2.53e+03	2.60e+03	5.06e+02	9.03e+02	2.13e+03	1.59e+03	1.61e+03	3.33e+02
F82022	2022	20	1.24e+04	2.62e+11	5.55e+09	1.97e+10	4.29e+10	1.24e+04	1.39e+09	6.14e+07	1.88e+08	3.16e+08	1.24e+04	2.27e+07	1.23e+06	3.37e+06	4.70e+06
F92022	2022	20	7.78e+02	1.98e+03	1.41e+03	1.37e+03	2.71e+02	6.53e+02	1.21e+03	9.23e+02	9.06e+02	1.30e+02	5.24e+02	9.33e+02	6.77e+02	6.86e+02	8.14e+01
F102022	2022	20	7.99e+02	5.19e+03	4.50e+03	3.76e+03	1.44e+03	6.77e+02	3.53e+03	1.38e+03	1.56e+03	6.69e+02	6.01e+02	1.25e+03	8.30e+02	8.51e+02	1.36e+02
F112022	2022	20	1.60e+03	4.80e+03	3.40e+03	3.37e+03	6.41e+02	1.12e+03	2.67e+03	2.24e+03	2.18e+03	2.70e+02	4.06e+02	2.05e+03	1.53e+03	1.47e+03	3.70e+02
F122022	2022	20	3.14e+02	4.80e+02	4.26e+02	4.21e+02	3.93e+01	3.14e+02	3.77e+02	3.42e+02	3.44e+02	1.38e+01	2.76e+02	3.33e+02	3.12e+02	3.10e+02	1.19e+01